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Interactions Between Parents and Infants, and Infants-father Separation in the Common Marmoset (*Callithrix jacchus*)

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ABSTRACT. The New World monkey, *Callithrix jacchus*, diversely from most Old World primates, shares infant care among all family members. This paper intends to describe a pattern of infant care in the common marmoset, and to verify if the infant-father separation induces a depression reaction, such as found in other separation studies. In Experiment 1, 11 families (mother, father, and twin newborn infants) of *Callithrix jacchus* were observed from 1st to 36th days of infants' life. It was noted that infants were carried most of the time in the beginning of life, and that from the 4th day on this care came mostly from the father. In Experiment 2, nine families were divided in three separation groups (15, 30, and 45 days of infants' life). Each group had a control and two experimental families. Removal of the father was done for three days, beginning at the indicated infants' ages. Results showed that upon removal of the father, the mother assumes the infants' care, and no depression reaction appeared.

Key Words: *Callithrix jacchus*; Infant-father separation; Family relationships; Infant care; New World monkeys.

INTRODUCTION

The study of the relationship between infants and their caregivers among nonhuman primates has been most valuable for the understanding of issues such as attachment and loss, infant socialization and development, and patterns of care in different species and situations (DIENSKE et al., 1982; DOLHINOW & MURPHY, 1982; GUBERNICK, 1981; HINDE, 1974; LAHIRI & SOUTHWICK, 1966; MCKENNA, 1979; MILLER & NADLER, 1981; ROSENBLUM & PAULLY, 1984). Since the pioneer studies of HARLOW (1958, 1959) much has been done on this field, and the effects of disturbances on that relationship, such as the removal of mother and peers, has been of interest to many researchers (MINEKA & SUOMI, 1978; ROSENBLUM & PLIMPTON, 1981). As described by SPITZ (1946) and BOWLBY (1973) in human children the response to separation is characterized by an initial extensive agitation, which BOWLBY called the protest stage, followed by despair and social withdrawal, and a detachment stage, which occurs after reunion. This response was first replicated in rhesus monkeys by SEAY et al. (1962), and since then by many other researchers in this and other species (MINEKA & SUOMI, 1978; ROSENBLUM & PLIMPTON, 1981), demonstrating that this response is not specific to human beings, at least in relation to the protest and despair stages (MINEKA & SUOMI, 1978; ROSENBLUM & PLIMPTON, 1981).

Most of these studies, though, have used Old World monkeys as subjects: animals in which infant contact characteristically occurs primarily if not exclusively with the mother. More

recently, however, workers have begun to study infant-caregiver relationships in New World monkeys, especially the Callitrichidae. These animals offer several advantages, in that they are small, easily bred in captivity, and long-living. They can also be housed in captivity in social groupings approximately to those found in the wild (EPPLE, 1970; STELLAR, 1960; STEVENSON, 1977). Of major interest is the fact that all family members participate in the care of the young (EPPLE, 1970, 1975; HOAGE, 1981; INGRAM, 1977; KLEIMAN, 1977; LOCKE-HAYDON & CHALMERS, 1983; STEVENSON, 1977). In common marmosets (*Callithrix jacchus*), in particular, in which both mother and father carry their twin infants extensively during the first weeks of life, it has been argued that sharing of parental duties is essential since the mother could not on her own successfully rear both of her twins (INGRAM, 1977). INGRAM also argues that the father's participation in infant care provides increased protection for the family against predators and liberates the female for foraging and enhanced milk production.

Of course, such patterns of infant care also modify the attachment value of each caregiver, given that the role of the mother is not so emphasized as in Old World primate species, and that the male and older siblings assume a new prominence in the care of the young (GUBERNICK, 1981; INGRAM, 1977; LOCKE-HAYDON & CHALMERS, 1983). LOCKE-HAYDON (1984) suggests that the distribution of an infant's time and activities amongst different caregivers will be determined by the balance between the relative amounts of care-seeking by the infant and caregiving by other members of the family at a certain infant age. This model allows one to explain differences induced by alterations in the composition and dynamics of the family, and events such as the promotion of the infant's independence (LOCKE-HAYDON & CHALMERS, 1983).

The work reported in this paper intends to describe family relationships in groups of common marmosets consisting of a reproductive pair and twin infants during the first month of life. We also report on the alterations brought about the removal of the father in three different life periods of the infants.

METHOD

EXPERIMENT 1

Subjects

Eleven families of *Callithrix jacchus*, from the Center of Primatology of Universidade Federal do Rio Grande do Norte, each consisting of a reproductive pair and twin infants were observed. Each family lived in a cage ($60 \times 75 \times 100$ cm), in a room with controlled temperature and a 12-hr light/dark cycle. Animals were fed twice a day, at 9:00 a.m. and 3:00 p.m., one meal consisting of fresh fruits and the other of a proteic mixture. Animals were not manipulated, except for washing of the cages every three days, when they were removed for a few hours.

Procedure

Families were observed in their home cages, through a one-way window, three days a week, for a 30-min period each day. Observations were made during the first month of infants' life. Both infants were observed in each family, and the data represent averages of all infants.

The following behaviors were observed: (1) the time the infants were being carried by

either the mother or the father, called time on; the time they were carried by the mother, time on *m*; and by the father, time on *p*. It was also noted if the infant was being carried on a ventral or dorsal position; (2) the time the infants were not being carried, time off; (3) the time the infants were in physical contact with each other, measured only during time off; (4) behaviors performed by mother or father to remove infants they are carrying, such as "rubbing off" onto the bars, biting, or pushing. It was noted the duration of such episodes of rejection; and (5) infants' active and non-active behaviors, which included: *active*: deambulation, when the youngs performed any kind of locomotor behavior; social play, in the form of rough-and-tumble (animals grasped or clung to each other and performed whole body rolling and batting), chase (animals chased each other or ran back and forth with each other); exercise play, when the animal supported itself by two limbs or legs, e.g., when hanging from wire by the hind limbs; and exploratory behavior, when the infants displayed manipulatory behavior, touching, chewing, sniffing, and/or licking any object (definitions after CHALMERS & LOCKE-HAYDON, 1981), and *non-active*: stationary posture, when the animal sat, stood, lay, or held to the wire mesh, by itself or in physical contact with twin or an adult.

EXPERIMENT 2

Subjects

Nine families of *Callithrix jacchus*, each consisting of a reproductive pair and twin newborn infants were used in this experiment. These families belonged to the Center of Primatology of Universidade Federal do Rio Grande do Norte.

The families were divided in three separation groups: (1) father's removed when the infants were 15 days old; (2) father's removed when the infants were 30 days old; and (3) father's removed when the infants were 45 days old.

In each group there were two separation and one control families.

Procedure

Procedure and behaviors observed were the same as those of Experiment 1. The periods of separation were chosen arbitrarily, aiming at different phases of the infants relationship, with its father and mother.

Two days before (pre-separation period), three days of separation (separation period), and two days after separation (post-separation period), the families were observed for two 30-min periods daily, in the morning and in the afternoon.

Removal of the father to another room was done immediately before the observation session on the first day of the separation period; mother and infants remained in the home cage.

RESULTS

EXPERIMENT 1

Results showed that the relationship between parents and their infants changes from a complete dependence to a partial independence during the first month of life.

Figure 1 presents the mean percentage of time on (mother plus father) and off from 1st to

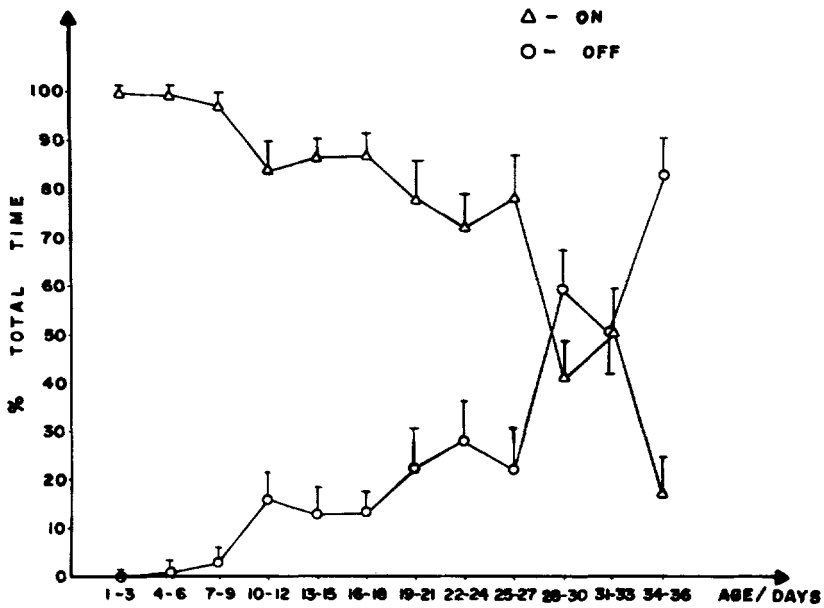


Fig. 1. Percentage of time infants were being carried (time on) or were not being carried (time off) from 1st to 36th day. Data are presented in averages for 3-day periods, and vertical lines represent SE.

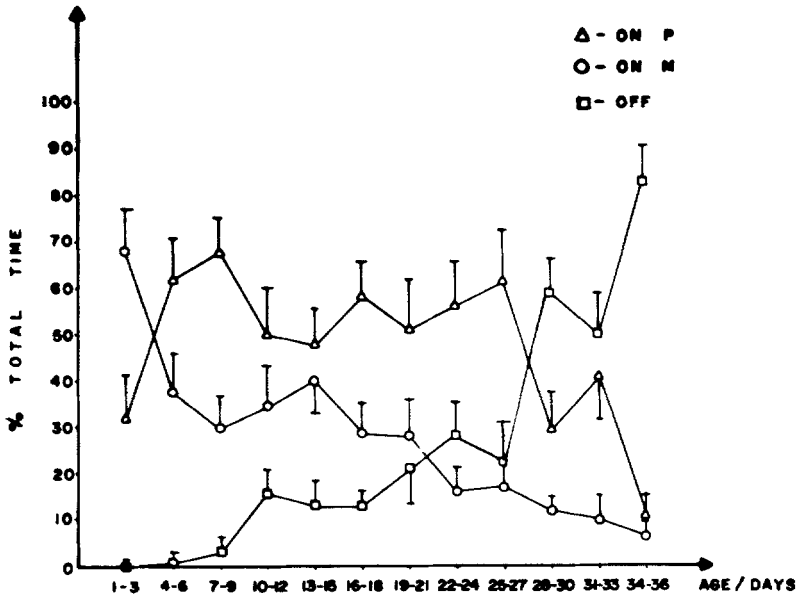


Fig. 2. Percentage of time infants were being carried by the mother (time on *m*), by the father (time on *p*) and were not being carried (time off), from 1st to 36th day, Data are presented in averages for 3-day periods, and vertical lines represent SE.

36th days of life; it shows that until the 9th day the infants spend almost all the time on the adults (range 97–100%). From day 10 to 27 there is a decrease in time on (72–87%), and the infants spend some time off the carrier (14–27%). This trend is even more clear during the 4th week when, for the first time, values of time on and off reverse, with time on showing lower percentages than time off. The decrease in time on and increase in time off get progressively more evident, with extreme values in days 34 to 36 (on 17%; off 83%).

How the male and female divide their time in carrying the infants is shown in Figure 2, which presents the mean percentage of time the infants are carried by mother and father (time on *m* and on *p*), and time off. As it can be seen, during the first days care by the mother is prevalent over care by the father. From the 4th day on the male takes over most of the carrying. Time on *m* shows some variability, but remains always lower than time on *p*. Time on *p* is specially high from 4th to 9th day (range 62–73%), and from 16th to 27th day (51–66%). From then on, time off is prevalent. It is interesting to note, though, that time on *m* is already lower than time off from the 22nd day onwards.

The main factor controlling the amount of time infants spend on or off seems to be rejection by the adults. Figure 3 shows the sum of the duration of the rejection episodes for mother and father. It is clear that for most of this period mothers rejected their infants more than did fathers. This could account for the lower amount of time that infants spent on mothers than fathers.

Three peaks of rejection by the mother were seen on the 7th to 9th, 13th to 15th, and 19th to 24th days. The 1st and 3rd coincide with an increase in time on *p*, and at least on two occasions it foretells a decrease in time on *m*. In relation to the male, the peaks for rejection were noted from 13th to 18th, and from 22nd to 24th days. They coincide with two peaks for the mother, and one of them is associated with an increase in time on *p*, suggesting that the

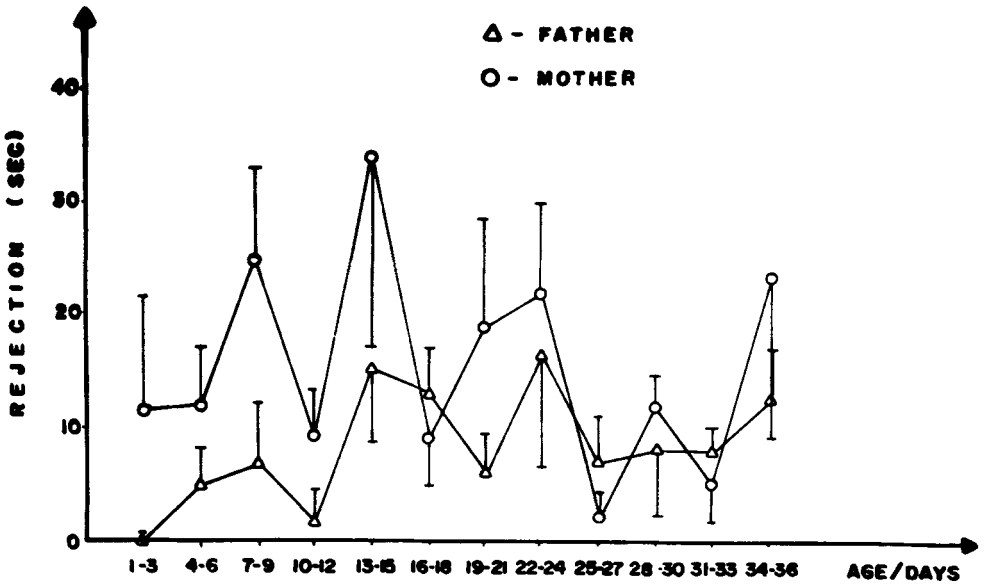


Fig. 3. Sum of the duration of the episodes of rejection (in sec) by mother and father, from 1st to 36th day. Data are presented in averages for 3-day periods, and vertical lines represent SE.

Table 1. Regression analysis (forward) of the variables age, time on *p*, and time on *m*, in relation to time off.

Variable	Correlation with time off	Percentage of variability explained	*f* (Fisher and Snedecor)
Age	0.82	66.8	68.28*
Time on *p*	−0.67	88.1[1)]	56.19*
Time on *m*	−0.44	69.6[1)]	3.11

*$p \leq 0.01$. 1) Percentage of variability explained by the variable, plus the one from age.

increase in rejection is due to an increase in approaching and seeking by the infants. The father's level of rejection, nevertheless, are always lower than those of the mother, even at the peaks. But Student's test shows a significant difference between rejection of mother and father only on the 7–12 days period ($p<0.01$). Noteworthy is the decrease in rejection by both mother and father from the 28th day on, coinciding with an increase in time off. The hypothesis that time on is controlled by parental rejection is reinforced by a significant negative correlation between time on and rejection (Pearson, $\rho = 0.77$, $p<0.001$).

Table 1 shows a regression analysis (forward) used to verify the influence of the variables age, time on *p*, and time on *m*, on the dependent variable time off. The correlation matrix showed that the most influential variable on time off was age, followed by time on *p*: age explains 66.8% of the variability of time off, which added to time on *p*, explains 88.1%. When added to time on *m* however it explains only 69.6%. In other words, variability in time off can be explained, first of all, by age and this in turn implies that growth of the infants, and in particular their increase in weight, and development of motor and social abilities, induces an increase in time off. Secondly, variability in time off may be explained by the amount of time the infants are carried by the father. This suggests that the greater responsibility in promoting the independence of infants in the first month of life belongs to the father when compared to the mother. But, it is important to remember that this is a very specific family situation (one reproductive pair and twin infants), occurring only on first breeding.

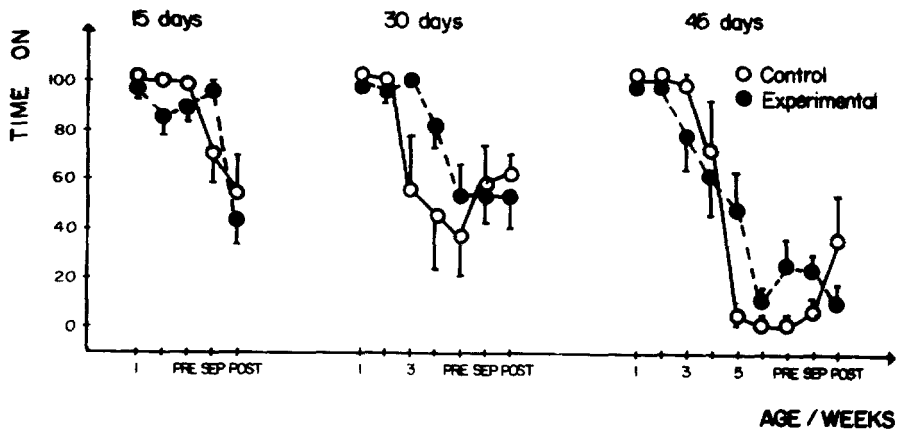


Fig. 4. Percentage of time infants were carried (time on) for experimental and control families, at 15, 30, and 45 days. Data are presented in averages for weeks, and for the pre-separation, separation, and post-separation periods. Vertical lines represent SE.

The presence of siblings may alter this situation, as it is pointed out by other authors (INGRAM, 1977; LOCKE-HAYDON & CHALMERS, 1983).

EXPERIMENT 2

Results showed that removal of the father did not bring about significant alterations in many measures of family relationships. Figure 4, which presents the mean percentage of time on for separation and control families shows that both curves are very similar. ANOVA for the whole period of observation did not show any significant differences between the groups. In other words, removal of the father did not bring about differences in the time the infants are carried, time on being equivalent for separation and control families for the whole period observed.

Figure 5 shows the mean percentage of time on *m* for separation and control families at 15, 30, and 45 days. It can be noted that the percentages are always higher for experimental families on the separation period, and more remarkably higher at 15, than at 30, than at 45 days (Mann-Whitney, separation and control families, 15 days, $p = 0.013$, 30 days, $p = 0.021$, 45 days, $p = 0.032$). This may be explained by the greater dependence of the infants at the beginning of life, when they spend more time on .These data suggest that the distribution of responsibilities on care of the infants is balanced to maintain a certain level of care, making one of the parents assume the other's function in its absence. Figure 6 shows the mean percentage of time on *p* for control and separation families at 15, 30, and 45 days. Though highly variable, time on *p* shows, just as in Experiment 1, high percentages at the beginning of life getting progressively lower. Noteworthy is the fact that the levels of time on *p* do not increase in the experimental families during the post-separation period, remaining quite similar to the control families (Mann-Whitney, non significant).

A comparison of time on *m* and time on *p* of experimental families (prior to separation) to values of Experiment 1 families is present in Table 2. The families from Experiment 1 showed a higher time on *p* than on *m* from 2nd to 5th weeks of infants life. Thirty and

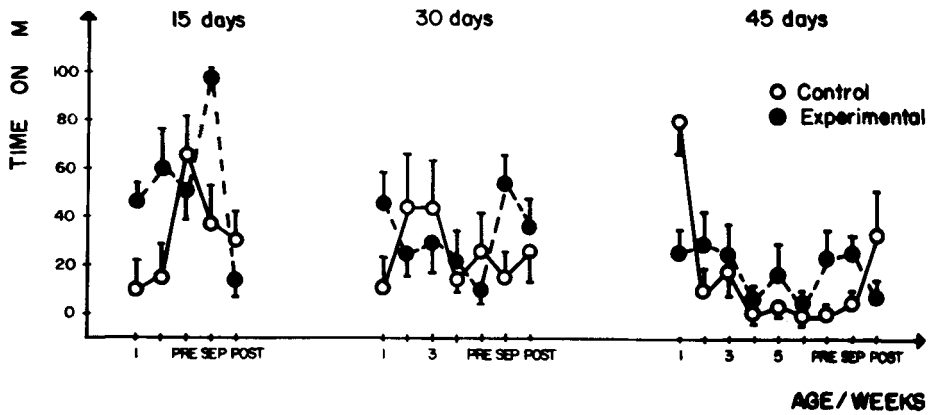


Fig. 5. Percentage of time infants were carried by their mothers (time on *m*) for experimental and control families at 15, 30, and 45 days. Data are presented in averages for weeks, and for the pre-separation, separation, and post-separation periods. Vertical lines represent SE.

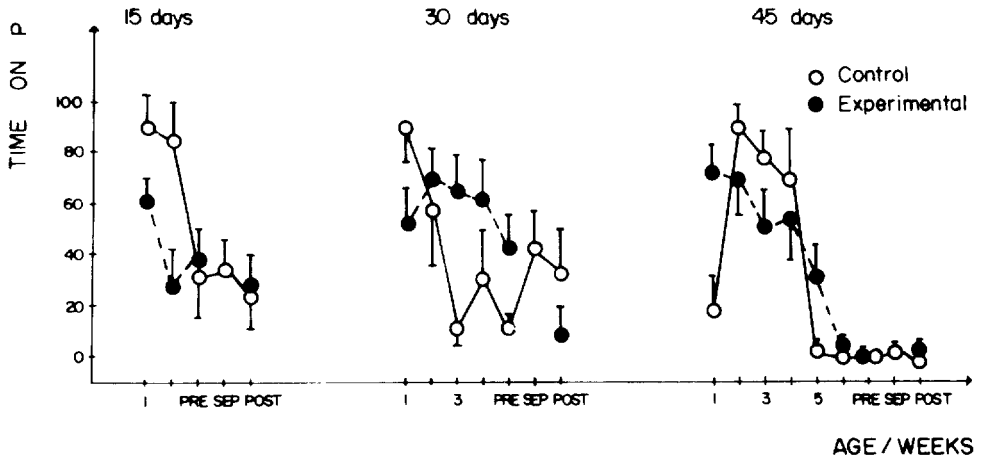


Fig. 6. Percentage of time infants were carried by their fathers (time on *p*) for experimental and control families, at 15, 30, and 45 days. Data are presented in averages for weeks and for the pre-separation, separation, and post-separation periods. Vertical lines represent SE.

Table 2. Mean time on *p* and on *m* for Experiment 1 and for 15, 30, and 45 days separation families.

	1st experiment		Sept. (15 days)		Sept. (30 days)		Sept. (45 days)	
Week	On *p*	On *m*	On *p*	On *m*	On *p*	On *m*	On *p*	On *m*
1st	49.5	49.8	61.6	36.7	44.0	54.2	68.4	31.6
	(5.3)	(5.3)	(6.4)	(6.3)	(11.8)	(11.6)	(10.9)	(10.9)
2nd	60.4	31.5**	26.0	60.1	70.6	25.7**	70.6	29.3
	(5.0)	(4.3)	(12.2)	(15.5)	(9.5)	(7.9)	(10.9)	(10.9)
3rd	54.2	29.3**	38.2	51.1[2)]	69.7	29.9	51.1	25.5
	(5.4)	(4.3)	(10.5)	(9.5)	(11.4)	(11.5)	(12.2)	(10.1)
4th	48.0	15.7**	—	—	53.6	27.0	54.6	6.7**
	(5.8)	(3.9)			(14.4)	(12.0)	(12.1)	(2.6)
5th	28.9	13.3**	—	—	42.8	10.0**[2)]	31.5	16.7
	(5.1)	(2.8)			(9.6)	(3.2)	(10.3)	(9.6)
6th	13.7	3.8	—	—	—	—	4.6	7.5
	(6.9)	(1.8)					(1.7)	(4.0)
7th	—	—	—	—	—	—	0.3	24.6*[2)]
							(0.2)	(9.0)

1) Numbers inside brackets represent one SEM; 2) pre-separation period. $*p \leq 0.05$; $**p \leq 0.01$ (*t* test for non-independent groups).

45 days separation families showed a higher time on *p* on almost all the weeks observed, showing a pattern similar to Experiment 1 families, although not all differennces being significant. Fifteen days families did not present a pattern similar to that presented by the families from the 1st Experiment, time on *p* showing higher values than time on *m* only on the 1st week.

Figure 7 shows time of rejection in seconds, for experimental and control groups at 15, 30, and 45 days. Noteworthy is the lower rejection time for experimental families during the separation periods, suggesting that the mother, once the male has been removed, not only carries the infants for a longer time, but is more tolerant with them. Mann-Whitney test, however, shows a significant difference only for the 15 days families ($p = 0.02$).

Figure 8 shows the percentage of time off which the infants spent in physical contact with each other. ANOVA did not show any differences in time of contact between separation and

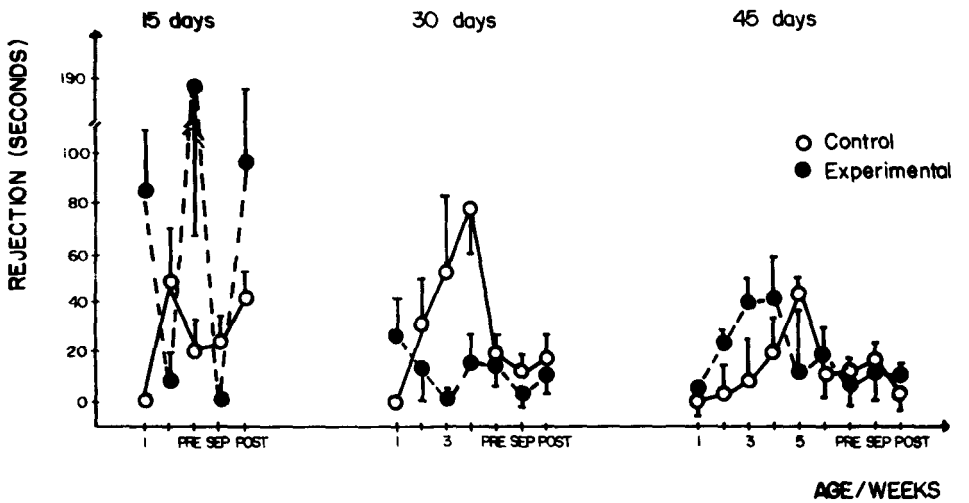


Fig. 7. Sum of the duration of the episodes of rejection (in sec) by mother and father for experimental and control families at 15, 30, and 45 days. Data are presented in averages for weeks, and for the pre-separation, separation, and post-separation periods. Vertical lines represent SE.

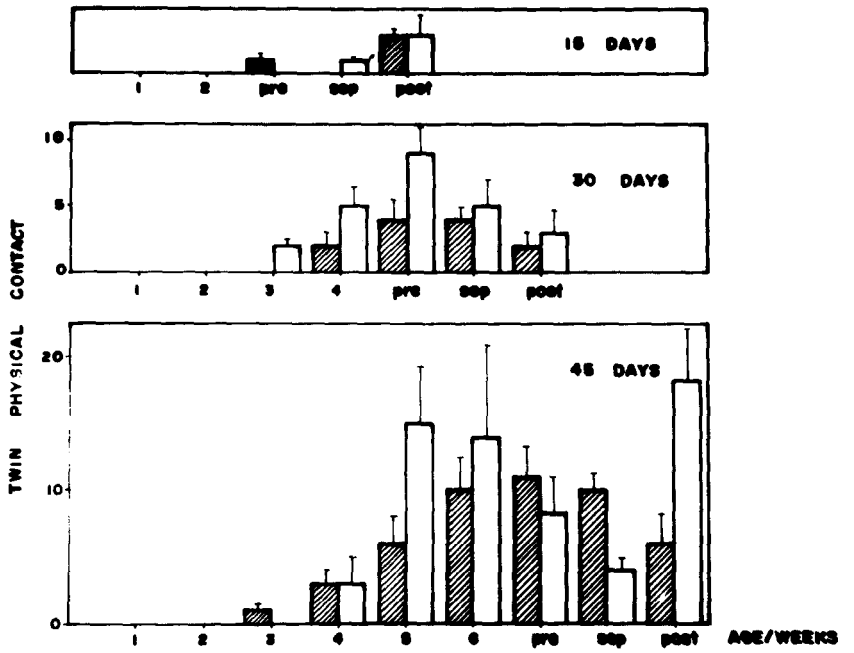


Fig. 8. Percentage of physical contact between twins during time off in experimental (hatched bars) and control (open bars) families, at 15, 30, and 45 days. Data are presented in averages for weeks, and for the pre-separation, separation, and post-separation periods. Vertical lines represent SE.

control families for the whole period observed. Although contact have a tendency to decrease for separation families during the separation period, Mann-Whitney test did not show any significant differences.

Analysis of the activity and non-activity data on the periods off suggests that there is no

difference between separation and control families. But, this analysis is difficult, since activity behaviors occur infrequently up to 45 days.

DISCUSSION

Our data on parent/infant relationship are in general agreement with those from INGRAM (1977) and LOCKE-HAYDON and CHALMERS (1983), but comparisons are not always easy, since we observed in more detail a much smaller period of time.

These authors found the decrease in time on to minimal levels by 6 weeks, interactions between twins from 3rd week on, and, as in our animals, a decrease in time on due primarily to rejection. LOCKE-HAYDON and CHALMERS (1983), though, call attention to a difference between their results and those of INGRAM's (1977) relating to the mother's level of carrying and rejecting the infants. Our results, in agreement with INGRAM's, show that the mother carries less and rejects the infants more than the father. It seems that the mother limits her care primarily to nursing, leaving contact confort to the male, and later to the twins, as the results of contact between infants show. Our results also show a peak of rejection (22nd to 24th days) for both mother and father just prior to the most striking decrease in time on (4th week), when values of time on and off reverse. From then on the rejection level is very low.

This pattern of care is fairly well reproduced in the separation experiment, in 30 and 45 days separation families, although a higher variability and a smaller number of subjects did not always resulted in statistically significant differences. But 15 days separation families showed a different pattern for mother and father time sharing of caregiving. Without significant differences, time on *m* was higher than time on *p* on 2nd and 3rd weeks. These results are due basically to one of the two families, once, in the other, time on *p* is higher than time on *m* from 1st to 3rd weeks. The occurrence of these abnormal patterns has been discussed in another paper (YAMAMOTO et al., in prep.), and although the reasons for their occurrence are not clear, the removal of the father in those families seems to influence the mother's caregiving behavior. That means, even when the conditions of caregiving sharing are not usual, the absence of the father exacerbates the caregiving behavior of the mother, inducing a greater compensation.

In relation to separation two points are noteworthy: first, that once the male who normally spends more time than the mother carrying the infants, has been removed, the female takes over the care; second no depression reactions appeared in the infants.

The first point relates to the mother's role and amount of care. LOCKE-HAYDON and CHALMERS' (1983), INGRAM's (1977), and our data are in complete agreement as to the distribution of care in the first 4 weeks of life: in this period care comes mostly from the father. But INGRAM reports that the mother is the most influenced by the presence of siblings, limiting care almost exclusively to nursing when there are other carriers available. Our results show that she can also be influenced the other way around: if she diminishes care when there are others available, she increases it when she is the only carrier. So, our hypothesis is that she is the one who determines the balance of infant care in the family, offering care whenever necessary, but withholding it if other carriers are available.

In a normal, undisturbed situation, she may leave infant care to the other members of the family, keeping her time for foraging and milk production. It may be vital to the mother that she reduces the care of her infants to a minimum, since she is likely to be pregnant, and therefore bearing simultaneously the costs of gestation (KLEIMAN, 1977; HUBRECHT, 1984).

It seems, however that she is the one who assesses the amount of care the infants receive from the various carriers, balancing it when necessary. LOCKE-HAYDON (1984) suggests a model of caregiving/care-seeking balance in *Callithrix jacchus*, from which she derives various predictions, one of them being that should levels of caregiving be reduced from one or more caregivers, compensation from other caregivers should not occur. This is very well demonstrated in her paper, regarding drugged father and siblings. The fact that compensation does occur with our families suggests that the presence or absence of the caregiver is a fundamental variable. Modification in the behavior of a caregiver has little influence on the caregiving behavior of other members of the family, but the absence of one of them induces the others (or, at least, the mother) to make good the amount of care normally offered by the missing member.

The second point refers to the depression reaction. Since the classical SEAY et al.'s study (1982), the protest despair reaction to separation has been replicated many times, making one believe that this is a universal response among primates. KAUFMAN and ROSENBLUM (1969) and KAUFMAN and STYNES (1978), though, throw a new light on this problem, showing that the social environment and specially the presence of substitute caregivers may prevent the development of a depressive reaction (MINEKA & SUOMI, 1978; ROSENBLUM & PLIMPTON, 1981; DOLHINOW, 1980).

ROSENBLUM and PLIMPTON (1981) point out that although all separation studies using nonhuman primates until the present time have shown what has been termed a "protest reaction," not all species or all individuals within a species subsequently show a despair phase. In our study, when one considers that other familiar caregivers were available, who were willing to care for the infants for at least the same amount of time as they were cared for in normal conditions, it is not surprising that infants did not show a depressive reaction.

The preseparation history of the infants and caregivers has been pointed out as an important source of variation in the response to maternal separation, anxiously attached infants showing a tendency to more severe depression reactions than securely attached infants (MINEKA & SUOMI, 1978). KAUFMAN and ROSENBLUM (1969) have shown that the more permissive the mother and the social environment the less likely is the infant to exhibit depressive reactions to separation. Accordingly, the *Callithrix jacchus* social environment tends to be very permissive once they are a kinship group, and from very early in life the infants interact with other members of the family besides the mother and are encouraged to explore the environment. The infant's twin probably also plays an important role in providing social support, since the twins remain in close contact before, during and after separation, either on or off the carriers. ROSENBLUM and PLIMPTON (1981) suggest that a separated infant may be responding not only to the psychological loss but also to the disruption of the entire social network, whose key of access is the mother. That does not happen in our experiment with the marmoset, since the social network became impoverished but not totally disrupted.

Most surprising is the lack of an apparent protest reaction, which is exhibited by all separated nonhuman primate infants, varying in length and intensity. The typical reaction to the father's removal in our animals was the mother running to the infants to retrieve them, and remaining with both on for a time that varied from a few minutes to the whole period of observation. Sometimes this evidenced a striking change in the mother's behavior, passing from high levels of rejection and intolerance to the infants in the pre-separation period, to low or no rejection and an augmented receptivity to them, during the period of separation.

Several variables seem to be implicated in the infants response to separation, especially

in the lack of the protest phase. Casual observations in our laboratories show that in the presence of any disturbing events, the adults tend to retrieve the infants. If we regard the father's removal as a disturbing event, the retrieving of the infants by the mother seems to be an initial response to the threat represented by the technician who removes the animal, and not to separation *per se*. Probably later the removal is perceived and the three animals left in the cage will respond to it. In the view of ROSENBLUM and PLIMPTON (1981), the protest reaction appears as a closed genetic program, so it is a necessary consequence of separation. It seems to us that a quite different behavior evolved for *Callithrix jacchus* whose most adaptive reaction upon separation from a caregiver is to look for another of its multiple caregivers. Separation from the mother may result in a different reaction, once the infants depend on her for milk at least until 1.5 to 2 months.

The protest reaction, however, does appear in the marmoset on other occasions. We have observed high levels of vocalization and care-seeking in infants who have been rejected and left on the cage wire for long periods, around the 2nd and 3rd weeks of life. This is probably a characteristic reaction in this species, but not present when the infants are separated from the father.

Experiment 1 regression analysis showed the father's greater influence over the time infants spend off when compared to the mother. Maybe we could even say there is a preference for the father as a caregiver. Indeed, LOCKE-HAYDON (1984) indicates care-seeking preferences of the infant *Callithrix jacchus* for different caregivers that vary with age. It is reasonable to think that these may also vary upon changes on the social environment, especially a drastic one such as the removal of the father. That way, providing there is a certain amount of care to those infants, removal of one of the caregivers would have a greater effect upon the others (or, in our case, upon the mother) and not upon the infants. Indeed, the effect of removal of the father on the mother is striking. Quantitative results were described before: increase of time on mother and decrease of rejection in the three separation groups. Other changes in the mother's behavior were noted such as: an apparent decrease in mobility, an increase on active seeking of the infants and of tolerance to them, evidenced by milder episodes of rejection and occurrence of eating with the infants on, never observed before separation.

In conclusion, the analysis of caregiving behavior in *Callithrix jacchus* showed that the father has a greater influence over the time infants spend off. Nevertheless, the mother seems to be the one to determine the amount of caregiving in relation to the number of caregivers, compensating for others in their absence. The experimental removal of the father helps us to understand the caregiving behavior of mother and father in *Callithrix jacchus*, and points to the great flexibility of these roles at least in relation to the mother. The very special social organization of the Callitrichidae among nonhuman primates probably led to a differential reaction to separation, once care of the young is shared by the family group. Further questions remain, such as the reaction to the mother's removal, or the influence of different separation environments, such as the absence of mother or twin, or the presence of older siblings. More research must also be done on the mother's reaction to the father's removal, since it was not explored in great detail in this paper.

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